CS 215: Task 05: September 14, 2026

1. X and Y are independent random variables with MGFs $M_X(t)$ and $M_Y(t)$. Derive the MGF of

$$W = aX + bY + c.$$

Hence, find $E[W]$ and $\text{Var}(W)$.

Teaching-team notes. Use only the independence property mentioned in the slides (Aug 24-Slide 13).

Solution: The MGF of W is

$$\begin{aligned} M_W(t) &= E[e^{tW}] \\ &= E[e^{t(aX+bY+c)}] \\ &= e^{ct}E[e^{atX}e^{btY}] \\ &= e^{ct}M_X(at)M_Y(bt), \end{aligned}$$

where independence of X and Y is used. Hence,

$$\boxed{M_W(t) = e^{ct}M_X(at)M_Y(bt)}.$$

Differentiating and evaluating at $t = 0$, using $M_X(0) = M_Y(0) = 1$,

$$M'_W(0) = aM'_X(0) + bM'_Y(0) + c,$$

so

$$\boxed{E[W] = aE[X] + bE[Y] + c}.$$

Similarly, using

$$\text{Var}(W) = M''_W(0) - [M'_W(0)]^2,$$

we obtain

$$\boxed{\text{Var}(W) = a^2\,\text{Var}(X) + b^2\,\text{Var}(Y)}.$$

2. $X_1, X_2, X_3 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. Let $T = X_{(2)}$ denote their median, i.e., the middle value when $X_{(1)} \le X_{(2)} \le X_{(3)}$ are in order. What is the pdf of T?

Solution: Let

$$F(t) = P(X_i \le t) = \Phi\left(\frac{t-\mu}{\sigma}\right),$$

where Φ is the standard normal CDF.

For the median to be at most t, at least two of the three observations must be at most t. Hence,

$$\begin{aligned} F_T(t) &= P(T \le t) \\ &= \binom{3}{2}F(t)^2[1 - F(t)] + F(t)^3 \\ &= 3F(t)^2 - 2F(t)^3. \end{aligned}$$

Thus,

$$\boxed{F_T(t) = 3F(t)^2 - 2F(t)^3}.$$

Differentiating,

$$f_T(t) = 6F(t)[1 - F(t)]f(t),$$

where

$$f(t) = \frac{1}{\sigma}\phi\left(\frac{t-\mu}{\sigma}\right).$$

Therefore,

$$\boxed{f_T(t) = \frac{6}{\sigma}\Phi\left(\frac{t-\mu}{\sigma}\right)\left[1 - \Phi\left(\frac{t-\mu}{\sigma}\right)\right]\phi\left(\frac{t-\mu}{\sigma}\right)}.$$

3. Let

$$X \sim \text{Bin}(n, p), \qquad Y \sim \text{Bin}(m, p)$$

be independent.

(a) Show that the conditional distribution of X, given $X + Y = k$, (i) is hypergeometric, and (ii) identify its parameters.

Teaching-team notes. Notation: $Z \sim \text{Hypergeometric}(N, W, K)$ if Z is the number of successes in a sample of size W drawn without replacement from a population of N objects containing K successes.

Solution: Note that, as mentioned in class, $X + Y \sim \text{Bin}(n + m, p)$. For x such that

$$\max(0, k - m) \le x \le \min(n, k),$$

we have

$$\begin{aligned} P(X = x \mid X + Y = k) &= \frac{P(X = x, Y = k - x)}{P(X + Y = k)} \\ &= \frac{\binom{n}{x}p^x(1-p)^{n-x}\binom{m}{k-x}p^{k-x}(1-p)^{m-k+x}}{\binom{n+m}{k}p^k(1-p)^{n+m-k}} \\ &= \frac{\binom{n}{x}\binom{m}{k-x}}{\binom{n+m}{k}}. \end{aligned}$$

Therefore, per notation, with $N = n + m, W = k, K = n$

$$\boxed{X \mid (X + Y = k) \sim \text{Hypergeometric}(n + m, k, n)}.$$

(b) Explain in words, without further calculation, why the parameter p disappears after conditioning.

Solution: Since $X_1, \ldots, X_n, Y_1, \ldots, Y_m$ (the underlying trials) are iid Bernoulli(p), every arrangement of successes and failures among the $n+m$ trials with exactly k successes has the same probability, $p^k(1-p)^{n+m-k}$, regardless of which k positions are the successes. So, conditional on there being k successes in total, all $\binom{n+m}{k}$ arrangements are equally likely; this is exactly the

same as randomly choosing which k of the $n+m$ trials are "black balls," with no dependence on p – equivalent to drawing without replacement from an urn.

4. A random ordering is made of a population containing K successes among N objects. In that ordering, let X be the number of successes among the first n_1 positions and Y the number among the next n_2 positions, where $n_1 + n_2 \leq N$. Find $\mathrm{Cov}(X, Y)$.

   (a) Explain, without any calculation, the sign of the covariance.

   Solution: The covariance is **nonpositive** because sampling is without replacement: if the first portion contains more successes, fewer successes remain available for the second portion, and vice versa.

   (b) Find $E[X]$ and $E[Y]$.

   Solution: Let I_i be the indicator that the i-th position contains a success. As mentioned in class, each position has probability K/N of being a success. Therefore,

$$\boxed{E[X] = \frac{n_1 K}{N}}, \qquad \boxed{E[Y] = \frac{n_2 K}{N}}.$$

   (c) Find $\mathrm{Cov}(X, Y)$.

   Solution: Let I_i be the indicator that the i-th position contains a success. Then

$$X = \sum_{i=1}^{n_1} I_i, \qquad Y = \sum_{j=n_1+1}^{n_1+n_2} I_j.$$

   For two distinct positions $i \neq j$, $N > 1$,

$$P(I_i = 1, I_j = 1) = \frac{K}{N}\frac{K-1}{N-1}.$$

   Since

$$E[I_i] = E[I_j] = \frac{K}{N},$$

   we get

$$\begin{aligned}\mathrm{Cov}(I_i, I_j) &= E[I_i I_j] - E[I_i]E[I_j] \\ &= \frac{K(K-1)}{N(N-1)} - \frac{K^2}{N^2} \\ &= -\frac{K(N-K)}{N^2(N-1)}.\end{aligned}$$

   Every pair consisting of one position from the first portion and one from the second portion has this same covariance. There are $n_1 n_2$ such pairs. Hence,

$$\mathrm{Cov}(X, Y) = n_1 n_2 \,\mathrm{Cov}(I_i, I_j),$$

   so

$$\boxed{\mathrm{Cov}(X, Y) = -\frac{n_1 n_2 K(N-K)}{N^2(N-1)}}.$$

5. Let $X_1, X_2, \ldots, X_k$ be independent random variables such that

$$X_i \sim \text{Geom}(p_i), \qquad i = 1, \ldots, k,$$

where the geometric distribution is supported on $\{1, 2, 3, \ldots\}$, i.e.,

$$P(X_i = n) = (1 - p_i)^{n-1} p_i, \qquad 0 < p_i < 1.$$

Define

$$M = \min(X_1, X_2, \ldots, X_k).$$

(a) Determine the distribution of M and its parameter(s).

Solution: Let

$$q_i = 1 - p_i.$$

For $m = 1, 2, \ldots,$

$$\begin{aligned} P(M > m) &= P(X_1 > m, \ldots, X_k > m) \\ &= \prod_{i=1}^{k} P(X_i > m) \\ &= \prod_{i=1}^{k} q_i^m \\ &= \left(\prod_{i=1}^{k} q_i \right)^m. \end{aligned}$$

Hence,

$$\begin{aligned} P(M = m) &= P(M > m - 1) - P(M > m) \\ &= \left(\prod_{i=1}^{k} q_i \right)^{m-1} \left(1 - \prod_{i=1}^{k} q_i \right). \end{aligned}$$

Therefore,

$$\boxed{M \sim \text{Geom}\left(1 - \prod_{i=1}^{k} (1 - p_i) \right)}.$$

Thus, the parameter of M is

$$\boxed{p_M = 1 - \prod_{i=1}^{k} (1 - p_i)}.$$

(b) Find the exact probability that variable j achieves the minimum, i.e.,

$$P(X_j = M).$$

Solution: We have

$$\begin{aligned} P(X_j = M) &= \sum_{m=1}^{\infty} P(X_j = m,\ X_i \geq m \text{ for all } i \neq j) \\ &= \sum_{m=1}^{\infty} p_j q_j^{m-1} \prod_{i \neq j} q_i^{m-1} \\ &= p_j \sum_{m=1}^{\infty} \left(\prod_{i=1}^{k} q_i \right)^{m-1}. \end{aligned}$$

Using the geometric series,

$$\sum_{m=1}^{\infty} r^{m-1} = \frac{1}{1-r},$$

we obtain

$$\boxed{P(X_j = M) = \frac{p_j}{1 - \prod_{i=1}^{k}(1-p_i)}}.$$

This probability includes the possibility of a tie at the minimum.

6. Let $X \sim N(0, \sigma^2)$ and $Y = |X|$. Let $\Phi(x)$ and $\phi(x)$ be the CDF and PDF of a standard normal random variable (i.e., mean 0, variance 1).

   (a) Derive the CDF and PDF of Y in terms of $\Phi(x)$ and $\phi(x)$.

   Solution: For $y < 0$,

   $$F_Y(y) = P(Y \leq y) = 0.$$

   For $y \geq 0$,

   $$\begin{aligned} F_Y(y) &= P(|X| \leq y) \\ &= P(-y \leq X \leq y) \\ &= \Phi\left(\frac{y}{\sigma}\right) - \Phi\left(-\frac{y}{\sigma}\right) \\ &= 2\Phi\left(\frac{y}{\sigma}\right) - 1. \end{aligned}$$

   Hence,

   $$\boxed{F_Y(y) = \begin{cases} 0, & y < 0, \\ 2\Phi\left(\frac{y}{\sigma}\right) - 1, & y \geq 0. \end{cases}}$$

   Differentiating,

   $$\boxed{f_Y(y) = \frac{2}{\sigma}\phi\left(\frac{y}{\sigma}\right), \qquad y \geq 0,}$$

   and $f_Y(y) = 0$ for $y < 0$.

   (b) Compute $E[Y]$.

   Solution: To find the mean,

   $$E[Y] = E[|X|] = 2\int_0^{\infty} x \frac{1}{\sigma}\phi\left(\frac{x}{\sigma}\right) dx = \boxed{\sigma\sqrt{\frac{2}{\pi}}}.$$

7. In an election, each of 1000 voters independently votes for candidate A with probability 0.52. We want to determine the probability that A receives a strict (i.e. ties not permitted) majority of the votes.

   (a) Complete the one-liner `python` script to compute this probability.

   ```
   from scipy.stats import ????
   prob =  ????
   print(f"{prob:.4f}")
   ```

   Solution:

   ```
   from scipy.stats import binom
   prob =  1 - binom.cdf(500, 1000, 0.52)
   print(f"{prob:.4f}")
   ```

   Value comes out to 0.8914

   (b) Use the normal approximation, the table in the Appendix of the textbook, and appropriate approximations to hand-calculate the probability without using any electronic aid.

   Solution: Let $X \sim \text{Binomial}(n = 1000, p = 0.52)$ be the number of votes for candidate A.
   With an electronic aid, we have

   - Mean: $\mu = np = 1000 \times 0.52 = 520$
   - Variance: $\sigma^2 = np(1-p) = 1000 \times 0.52 \times 0.48 = 249.6 \implies \sigma \approx 15.80$

   Without an electronic aid, we approximate the variance to 250 which is slightly higher.
   A strict majority requires $X \geq 501$. Applying the normal approximation with continuity correction: With an electronic aid

$$\begin{aligned} P(X \geq 501) &\approx P(X \geq 500.5) \\ &= P\left(Z \geq \frac{500.5 - 520}{\sqrt{249.6}}\right) \\ &= P(Z \geq -1.234) \\ &= \Phi(1.234) \approx \mathbf{0.89} \end{aligned}$$

   Without an electronic aid

$$\begin{aligned} P(X \geq 501) &\approx P(X \geq 500.5) \\ &\approx P\left(Z \geq \frac{500.5 - 520}{\sqrt{250}}\right) \\ &\approx P(Z \geq -1.25) \\ &= \Phi(1.25) \approx \mathbf{0.89} \end{aligned}$$

   (c) Use the normal approximation and `python` to compute the probability. Write the one-liner for the code fragment.

   Solution:

   ```
   from scipy.stats import norm
   prob =  1 - norm.cdf(500.5, 1000*0.52, (1000*0.52*(1-0.52))**0.5)
   print(f"{prob:.4f}")
   ```

   Value comes to 0.8914

   (d) Are you able to obtain the probability correct to 2 decimal places? If so, what approximations did you use?

Solution: Yes. The normal approximation with continuity correction gives 0.89 to two decimal places. We also approximate the z-value using the normal table.

8. The annual salaries of physicians are normally distributed. It is known that (in some currency)

   - 25% earn less than 180,000, and
   - 30% earn more than 320,000.

   (a) Find the mean (μ) and standard deviation (σ) of the salary distribution.

   Solution: Let X denote the salary in thousands. From the given conditions:

$$P(X < 180) = 0.25 \implies \Phi\left(\frac{180-\mu}{\sigma}\right) = 0.25 \tag{1}$$

$$P(X > 320) = 0.30 \implies P(X \le 320) = 0.70 \implies \Phi\left(\frac{320-\mu}{\sigma}\right) = 0.70 \tag{2}$$

   Using standard normal critical values:

   - $z_{0.25} = -\Phi^{-1}(0.75) \approx -0.6745$
   - $z_{0.70} = \Phi^{-1}(0.70) \approx 0.5244$

   This yields the linear system:

$$\begin{aligned} 180 - \mu &= -0.6745\,\sigma \implies \mu - 0.6745\,\sigma = 180 \\ 320 - \mu &= 0.5244\,\sigma \implies \mu + 0.5244\,\sigma = 320 \end{aligned}$$

   Subtracting the first equation from the second:

$$(0.5244 + 0.6745)\,\sigma = 320 - 180 \implies 1.1989\,\sigma = 140$$

$$\sigma = \frac{140}{1.1989} \approx 116.774 \implies \sigma \approx 116{,}774$$

   Substituting σ back to find μ:

$$\mu = 180 + 0.6745(116.774) \approx 258.764 \implies \mu \approx 258{,}764$$

   (b) Find the fraction earning between 260,000 and 300,000.

   Solution: We evaluate $P(260 \le X \le 300)$:

$$z_1 = \frac{260 - 258.764}{116.774} = \frac{1.236}{116.774} \approx 0.0106$$

$$z_2 = \frac{300 - 258.764}{116.774} = \frac{41.236}{116.774} \approx 0.3531$$

   Using the standard normal CDF Φ:

$$\begin{aligned} P(260 \le X \le 300) &= \Phi(0.3531) - \Phi(0.0106) \\ &\approx 0.6380 - 0.5042 \\ &= \mathbf{0.1338} \quad (\approx 13.38\%) \end{aligned}$$